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Boundary slip in Newtonian liquids: a review of experimental studies

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Abstract

For several centuries fluid dynamics studies have relied upon the assumption that when a liquid flows over a solid surface, the liquid molecules adjacent to the solid are stationary relative to the solid. This no-slip boundary condition (BC) has been applied successfully to model many macroscopic experiments, but has no microscopic justification. In recent years there has been an increased interest in determining the appropriate BCs for the flow of Newtonian liquids in confined geometries, partly due to exciting developments in the fields of microfluidic and microelectromechanical devices and partly because new and more sophisticated measurement techniques are now available. An increasing number of research groups now dedicate great attention to the study of the flow of liquids at solid interfaces, and as a result a large number of experimental, computational and theoretical studies have appeared in the literature. We provide here a review of experimental studies regarding the phenomenon of slip of Newtonian liquids at solid interfaces. We dedicate particular attention to the effects that factors such as surface roughness, wettability and the presence of gaseous layers might have on the measured interfacial slip. We also discuss how future studies might improve our understanding of hydrodynamic BCs and enable us to actively control liquid slip.

(Some figures in this article are in colour only in the electronic version)

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Contents

	Page
1. Introduction	2861
2. Historical foundations	2862
3. Basic hydrodynamic theory	2864
3.1. Newtonian fluids	2864
3.2. The Navier–Stokes equations	2865
3.3. Solution for a spherical geometry	2866
3.3.1. Stokes equations	2866
3.3.2. The approximate solution for the sphere–flat geometry	2867
3.3.3. The exact solution for the sphere–flat geometry	2867
3.4. Boundary conditions	2868
3.5. The slip length model	2869
3.6. Flow through capillaries	2870
4. Experimental techniques for the investigation of BCs	2871
4.1. Techniques tracing the fluid flow near a boundary	2871
4.1.1. Fluorescent recovery after photobleaching	2871
4.1.2. Using tracers to follow liquid flow	2872
4.2. Techniques based on force or displacement measurement	2874
4.2.1. Surface forces apparatus	2874
4.2.2. Atomic force microscopy	2875
4.3. Capillary techniques	2876
4.4. Quartz crystal resonators	2877
4.5. Other techniques	2877
5. Factors affecting slip	2878
5.1. Surface wettability	2878
5.2. Surface roughness	2879
5.3. Shear rate	2881
5.4. Nanobubbles or gaseous films	2881
5.5. Other factors affecting slip	2883
5.5.1. Adsorption of species from solution	2883
5.5.2. Curvature	2884
5.5.3. Polarity of the solvent	2884
5.5.4. Viscosity	2884
6. A mechanism for liquid slip	2884
7. Outlook	2886
8. Conclusions	2887
Acknowledgments	2888
References	2888

1. Introduction

The dynamics of flow in confined geometries, such as in microfluidic devices, can be accurately described only if the physics of the flow at the interface between the fluid and the solid is thoroughly understood. One of the most important steps towards this understanding lie in determining the correct boundary conditions (BCs) for the system being studied. The choice of BCs for a fluid flowing over a solid surface has been heavily debated over the past two centuries, but a convincing conclusion is still lacking. Finding the correct BCs that apply to flow in confined geometries is not only of great interest to the scientific community, but would also improve our understanding of a number of industrial and technological processes, such as flow in porous media [1–3], electro-osmotic flow [4, 5], particle aggregation [6] or sedimentation [7], extrusion through dies [8] and lubrication [9–11]. It would also provide a fundamental advance in understanding the physics of flow in microfluidic devices [12, 13] and biological processes [14, 15].

One of the core concepts of fluid mechanics is the no-slip BC, i.e. the assumption that the layer of liquid next to a solid surface moves with the same velocity as the surface. The no-slip BC cannot be proved based on hydrodynamic considerations, but has been demonstrated in numerous macroscopic experiments. Its physical origins are believed to be mainly the trapping of liquid in pockets and crevices present on the solid surface and the attractive forces between the molecules of the solid and those of the liquid. The general acceptance of the no-slip BC is the result of a debate between eminent scientists over two centuries, and it now forms the basis for most continuum-based calculations.

In recent years the topic of the flow of Newtonian liquids in confined geometries has gained new interest, and numerous research papers have cast doubts on the universality of the no-slip BC, showing that under certain circumstances fluid slip might occur at the solid boundary. Some early experiments indicating slip, mostly involving the flow of liquids through thin lyophobic capillaries [16–19] (see also references in [20]), have received revived interest as a number of new experiments, performed with more modern techniques, have also shown evidence of boundary slip [21–26].

Several new methods of investigation of the BCs for Newtonian liquids flowing over solid surfaces are currently being employed in experimental, computational and theoretical work. Our aim here is to review the most important experimental advances in this field. Excellent reviews covering the theoretical work [27, 28] and work done with quartz crystal resonators have recently appeared in the literature [29]. Here we summarize the experimental findings, comment on the pros and cons of the employed techniques, discuss the influence of various factors on the results and propose new lines of research.

This review is divided into eight sections: section 2 reviews seminal work produced in the 18th, 19th and early 20th centuries and the long discussion on BCs that eventually ended with the universal acceptance of the no-slip BC. Section 3 provides a brief introduction to hydrodynamic theory. Section 4 reviews the various experimental techniques that have been employed to study BCs, with a discussion of their advantages and disadvantages and particular emphasis on the results of the last five years. Section 5 reports on the main factors that influence the occurrence and magnitude of slip: surface wettability, surface roughness, the presence of nanobubbles and gaseous films and other possible factors. Section 6 summarizes debated physical models to explain the slip phenomenon. Section 7 provides an outlook on the future of the field, discusses pertinent experimental issues and suggests approaches for evaluating the role of a range of experimentally accessible parameters in determining the slip condition. Finally section 8 provides final remarks and conclusions drawn from our overview of the subject.

2. Historical foundations

We present here a very brief overview of the extensive discussion on BCs conducted by several distinguished scientists between the 18th and the early 20th centuries (see also [30–32]). Three main alternative BCs have been proposed and investigated.

The no-slip BC (see figure 1(a)). Early experimentalists did not observe slip of liquids at solid boundaries. Bernoulli [33] first proposed that for a fluid flowing across a solid surface the fluid velocity at the solid surface must be the same as that of the surface and gradually vary when moving away from the surface into the fluid. Shortly afterwards, as stated by Stokes [34], Du Buat [35] drew a similar conclusion with respect to experiments of water flowing through pipes. Du Buat reported that when the mean velocity of the water was below a given value, the water near the surface of the pipe was at rest. As described by Stokes [34], Coulomb [36] claimed that his experiments showed that the fluid molecules in contact with an oscillating cylinder have the same velocity as the cylinder; molecules at a short distance into the bulk fluid have smaller velocity and at a distance of several millimetres from the cylinder the velocity reduces to zero.

The stagnant layer BC (see figure 1(b)). According to this model, a thin layer of liquid is firmly attached to the solid surface, and the rest of the bulk liquid slips over this layer. Girard [37] first proposed this BC in 1813 and stated that, if the solid is of the same material everywhere, the thin liquid layer has a constant thickness. Also the thickness of the layer should depend on the curvature of the wall, on the type of liquid and solid, and on the temperature. The attached liquid layer thickness would become zero for liquids which do not wet the solid, such as mercury in glass tubes, in which case the liquid should slip over the surface [38]. Prony [39] (see [34]) proposed similar ideas but assumed a thicker stagnant layer. It was generally assumed that wettability should have a strong influence on the BCs.

The partial slip BC (see figure 1(c)). Navier [40] first proposed (as stated by Stokes [34]) that a liquid may slip on a solid surface, and this slipping would be opposed by a frictional force proportional to the velocity of the fluid relative to the solid. He introduced the idea of ‘slip length’, which is nowadays the most commonly used concept to quantify the slip of a liquid at a solid surface [41–47]. The slip length, b , is the distance beyond the liquid/solid interface at which the liquid velocity extrapolates to zero (see figure 1(c)). The slip length b is related to the velocity of the liquid at the wall v_r by the expression:

$$v_r = b \frac{\partial v_b}{\partial z}, \quad (1)$$

where v_b is the velocity of the fluid in the bulk and z is the axis perpendicular to the wall, as defined in figure 1.

By the 1830s the true nature of the BC at the fluid–solid interface had become a very contentious topic. Stokes [34] reported that Poisson [48, 49] had considered the combination of two BCs, suggesting the application of Navier’s condition at the outer surface of a stagnant or quasi-solid layer (Girard’s condition). In the 1840s Stokes was commissioned by the Royal Academy of Science to investigate the true nature of the BCs. After a period of indecision between Bernoulli’s and Navier’s hypotheses, caused by disagreement between his calculations and experiments known to him, Stokes finally favoured the no-slip BC [34]. His motivation was that any discontinuity in velocity at an imaginary surface in a liquid medium would give rise to a tangential force that would immediately ‘destroy the finite relative motion of particles indefinitely close to each other, so as to render the motion continuous’. This argument was then extended to a surface where a solid and liquid meet. Indeed some of Stokes’ calculations gave results consistent with the no-slip BC, when applied to experiments on the flow of water

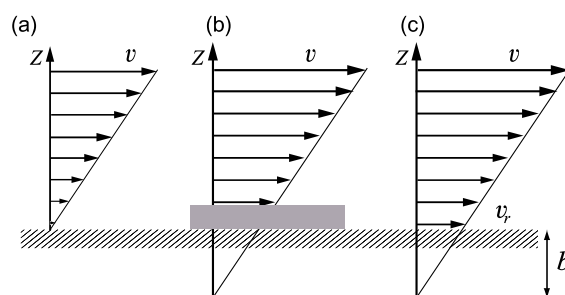


Figure 1. Schematic representation of (a) the no-slip BC, (b) the stagnant layer BC and (c) the partial slip BC. The length of the arrows is proportional to the velocity of the liquid. Under no-slip and stagnant layer BC the relative velocity, v , between the fluid and the solid wall is zero at the wall. In the stagnant layer BC, liquid slips on top of a thin layer of liquid attached to the solid wall (shaded region). In the slip BC, slip occurs at the wall, where $v = v_r > 0$. The extent of slip is characterized by the slip length b (see text).

and mercury through tubes of glass [50, 51]. However, several experiments reported flow behaviour that could not be described by the no-slip BC.

Reported in his memoirs on the movement of blood, Poiseuille [52] observed a stagnant layer of blood at the walls of blood-vessels. This led him to examine the flow of suspensions of liquids containing opaque bodies through glass tubes. He again observed stagnant layers at the walls, but much thinner ones than those described by Girard. This type of flow through a capillary is now commonly referred to as Poiseuille flow. Darcy [53] in his memoir on the flow of water through pipes substantially agreed with the stagnant layer hypothesis. Helmholtz and Piotrowski [54, 55] stated that the velocity of fluid in contact with a surface is not always equal to that of the surface itself, but a certain amount of actual slipping takes place. Experiments on the oscillation of spheres filled with liquid led them to the conclusion that though there might be no-slip in the case of water, ether or alcohol in contact with glass, there was a considerable slip for water in contact with a gilt surface (we note here that metallic surfaces are easily contaminated and in these experiments it is likely that these surfaces were hydrophobic). These measurements were later criticized by Maxwell, who thought that the employed technique lacked the necessary accuracy to distinguish such small slip lengths from a true no-slip BC [56]. Despite these efforts, the BC for fluid flow was still very obscure, due to a lack of experimental precision.

In the following years several voices supported the no-slip BC. Whetham [57] conducted experiments on the flow of water through silvered and copper tubes, the smallest of which had a diameter of 1 mm, and repeated some of Piotrowski's experiments [55]. He came to the conclusion that there was no evidence of slip, especially of the magnitude (about $400\ \mu\text{m}$) seen by Helmholtz. Couette [58], Maxwell [56] and Ladenburg [59] all independently arrived at the same conclusion.

By the 1900s it was generally accepted that slip, if it did exist, was too small to be observed. A quasi-solid layer of liquid (stagnant layer BC) was too thin, and a finite relative velocity between the liquid and solid at the surface (partial slip BC) was too small to make any observable difference in the results on the macroscopic scale. Importantly, it was concluded that the BC at the surface could depend on the velocity of the liquid [34, 35, 40] and the wettability [34, 37] of the surface.

More experiments with thin capillaries possibly showing slip were ascribed to the effect of the modification of surface tension by surfactants [60–62]. Agreement between observations and Stokes' and Oseen's theories of motion at small Reynolds numbers [41], as well as

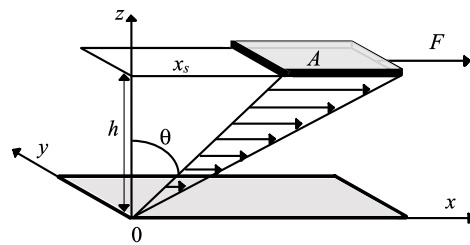


Figure 2. Flow of a liquid between two parallel plates distant h under an applied shear stress $\sigma = F \cdot A$.

agreement between Taylor's calculations and observations on the stability of flow between rotating cylinders [63], helped to justify the no-slip assumption on the macroscopic scale.

By the mid-20th century there had been no unambiguous determination of the BC at the liquid–solid interface; rather it had been unanimously accepted that even if slip occurred, it would have been detected only using experimental techniques with resolution far beyond that available at the time. Therefore the no-slip BC was widely employed for all practical applications. The no-slip BC became so accepted in the 20th century that most fluid dynamics textbooks failed to acknowledge that it was based on empirical knowledge (with some notable exceptions [30]). Towards the close of the 20th century a rapid development in the field of miniaturization and nanofabrication brought about a rapid growth of two new research areas, nano- and bio-technology. With the progression of these areas into the micro- and nanoscopic domains has come a demand for detailed understanding of physical phenomena on this small scale. The development of highly sensitive techniques for measuring interactions and displacements at the nanometre level now allows the investigation of BCs to be revisited with new experimental approaches.

3. Basic hydrodynamic theory

3.1. Newtonian fluids

Figure 2 illustrates a shearing experiment, where a material of thickness h is confined between two parallel plates. The lower plate is held stationary and the upper plate is pulled with a force F , acting in direction x over the area A for a certain amount of time. The plate exerts a shear stress, σ , i.e. a force per unit area, on the material immediately below it and causes a deformation or strain in it, $\gamma = \tan \theta$, where θ is defined as in figure 2. When the force is removed, the material will either recover its original shape (elastic behaviour) or remain in its new position (viscous behaviour: flow has occurred) or a partial recovery will occur (plastic behaviour). These three types of behaviour are characteristic of solids, liquids and plastic materials, respectively, but many materials exhibit a range of behaviour, depending on the time scale of the application of the stress and the time of observation [64, 65].

Newton described the flow of simple fluids (some liquids and gases) under a steady shear with the expression:

$$\sigma = \eta \cdot \frac{d\gamma}{dt}, \quad (2)$$

where η is the dynamic viscosity (expressed in pascal seconds) and $d\gamma/dt$ is the velocity gradient within the distance h and is called shear (or strain) rate. If the velocity of the upper

plate is v_x , then $d\gamma/dt = d(\tan \theta)/dt = h^{-1} \cdot dx_s/dt = v_x/h$, where h and x_s are defined in figure 2.

Liquids are defined as Newtonian when their behaviour is ideal, purely viscous, i.e. when their behaviour under shear can be described by equation (2). The viscosity of Newtonian liquids is simply a function of temperature and does not depend on shear rate, up to very high values of the shear rates. Also, in a more simplistic view, Newtonian liquids are normally made of low molecular weight molecules. Examples of liquids that in normal conditions behave in a Newtonian manner are water (at 20 °C, $\eta = 1.002$ mPa s), olive oil ($\eta = 99$ mPa s) and glycerol ($\eta = 2330$ mPa s). Non-Newtonian liquids include a wide category of liquids (polymers, dispersions etc) whose behaviour is more complex, and in particular whose viscosity depends critically on the shear rate.

A simplistic way to characterize interactions between molecules in a non-Newtonian liquid is to use a Lennard–Jones potential. One can then define a characteristic time τ_0 :

$$\tau_0 = \left[\frac{m\kappa^2}{\epsilon} \right]^{1/2}, \quad (3)$$

where m is the molecular mass and κ and ϵ are the characteristic length- and energy-scale of the molecules, respectively [66]. It has been shown in rheological studies that non-Newtonian behaviour commences when the shear rate approximately exceeds twice the molecular frequency scale τ_0^{-1} . For simple liquids such as water, this time scale is extremely short and the threshold shear rate for the onset of non-Newtonian behaviour is very high. For high molecular weight polymers, m and κ are orders of magnitude higher than for water; so the limiting shear rate for non-Newtonian behaviour is easily reached.

3.2. The Navier–Stokes equations

Two differential equations, referred to as the Navier–Stokes equations, describe most of the commonly occurring fluid flows. The derivation of these equations is an involved calculus exercise, and the interested reader is referred to textbooks on the subject [65–71]. In the following treatment we neglect the molecular nature of the fluid and treat the liquid as a continuum. Furthermore, we assume that the liquid is incompressible, which for water and aqueous solutions is a good approximation (the density of water changes by only 0.01% if the pressure increases from 1 to 2 atm). The incompressibility criterion, together with the principle of mass conservation, leads to the first of the two equations, referred to as the continuity equation, which in vectorial form is

$$\nabla \cdot \vec{v} = 0, \quad (4)$$

where $\vec{v}$ is the velocity vector describing the liquid flow. The continuity equation means that the mass of liquid flowing into any volume V over a period of time must be balanced by the amount flowing out. The second of the Navier–Stokes equations describing the velocity field in a Newtonian liquid is obtained by applying Newton’s second law of motion to a small block of fluid:

$$\rho \left(\frac{\partial \vec{v}}{\partial t} + \vec{v} \nabla \vec{v} \right) = \vec{F} - \nabla p + \eta \nabla^2 \vec{v}, \quad (5)$$

where ρ and η are the density and dynamic viscosity of the liquid, respectively, and p is the pressure. The terms on the right of the equal sign represent the total force per unit volume acting on the block of fluid. $\vec{F}$ is the total long-range force per unit volume and in most applications it is known, e.g. it may be the gravitational force $\vec{F} = \rho \cdot \vec{g}$. The terms $(-\nabla p + \eta \nabla^2 \vec{v})$ express the stress force per unit volume and $(\eta \nabla^2 \vec{v})$ represents the viscous force. From Newton’s

second law of motion, the term on the right of the equal sign can be equated to the mass times the acceleration per unit volume. Writing the acceleration in terms of the velocity field we obtain the term to the left of the equal sign in equation (5). The unknowns in equation (5) are the velocity and the pressure fields, which can be determined by solving the Navier–Stokes equations subject to particular BCs. Most commonly the layer of liquid next to the surface is assumed to move with the local velocity of the surface (no-slip BC). In the following sections we will present a few solutions of the Navier–Stokes equations.

3.3. Solution for a spherical geometry

The Navier–Stokes equations have been solved for several different geometries of the interfaces; here we will describe in detail only the solution for a spherical particle immersed in a fluid that extends infinitely in all directions. This is a useful example not only because it is one of the simplest geometries but also because it provides the theoretical background for several dynamic force measurements performed with the colloid probe atomic force microscope and the surface forces apparatus (SFA). In the following, vectorial quantities are presented only by their scalar part.

3.3.1. Stokes equations. The problem of the flow around a sphere immersed in a fluid is normally solved under the creeping flow approximation, i.e. neglecting inertial and gravitational terms [65, 70, 72]. Inertia terms are neglected because the flow field in this case has a very small length scale, i.e. in the neighbourhood of the particle the fluid velocity will vary with distance on a length scale of the order of the particle radius r . The velocity gradients will be of the order of (v/r) , where v and r are the particle velocity and radius, respectively. Hence the inertia term $\rho v \nabla v$ in equation (5) will be of the order of $\rho(v^2/r)$. Similarly the magnitude of the viscous force term $\eta \nabla^2 v$ will be of the order of $(\eta v/r^2)$. The ratio of these two factors is given by the Reynolds number Re :

$$\frac{|\text{inertia term}|}{|\text{viscous term}|} = \frac{|\rho v \nabla v|}{|\eta \nabla^2 v|} = \rho \frac{vl}{\eta} \equiv Re, \quad (6)$$

where l is the characteristic length scale of the flow, and for the sphere geometry it is $l = r$. For many macroscopic situations the Reynolds number is considerably larger than 1 and the flow is turbulent. For colloidal flows (or for flow in confined geometries) Re is usually small ($Re \ll 1$), as a consequence of the small particle radius (or small channels) and velocity, and the fluid flow around the particle is laminar. Thus for such microscopic flows we can neglect the inertial term $\rho v \cdot \nabla v$. In a sedimentation experiment the force F is approximately uniform in the neighbourhood of the particle, so it can be included in the pressure term that then becomes a modified pressure p' . Neglecting Brownian motion, the fluid velocity field is independent of time and equation (5) therefore reduces to the following:

$$\eta \nabla^2 v = \nabla p'. \quad (7)$$

Equations (4) and (7) are called the Stokes equations. They are commonly solved under the no-slip BC. Solving the Stokes equations for a spherical particle of radius r immersed in a viscous fluid of bulk dynamic viscosity η and moving with a velocity v , one obtains the drag force (Stokes formula):

$$F_d = 6\pi\eta r v, \quad (8)$$

where F_d is the magnitude of the drag force, which acts in a direction opposite to the direction of movement of the particle. Experiments confirm the approximate validity of this formula at low Reynolds number. On the other hand in a turbulent flow, the drag force is proportional to the square of the velocity.

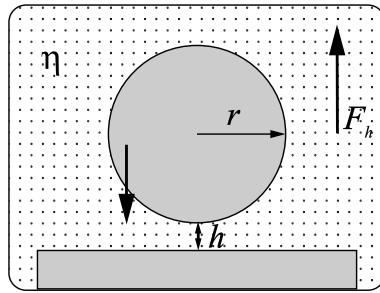


Figure 3. A sphere of radius r immersed in a liquid of viscosity η approaches a solid wall with a velocity v at a distance of closest approach h (with $h \ll r$). An opposite drainage force F_h arises.

3.3.2. The approximate solution for the sphere–flat geometry. In practice the drag force on an isolated sphere is not commonly measured, but the approach of a sphere to a flat wall (or an equivalent geometry) is studied. This geometry is in fact employed in dynamic force measurements performed recently with the SFA and atomic force microscopy (AFM) (see, e.g. [22,73–78]). These techniques measure directly the hydrodynamic force on a sphere approaching a flat surface, as shown in figure 3. Note that the cross cylinder geometry employed in the SFA (see figure 7) is equivalent to the sphere–flat geometry if the cylinders are of equal radius.

The problem of the flow of a liquid confined between two approaching surfaces was first treated by Reynolds in the creeping motion approximation [9]. Further assumptions made by Reynolds for the solution of this particular case are known as the lubrication approximation: in essence for small surface separations, h , the thickness of the liquid film separating the two surfaces is small compared with the radius of curvature r of each surface, and so locally the surfaces are nearly parallel. The flow of the liquid occurs mainly in the radial direction, and the derivatives of the velocity in the radial direction are negligible compared with those in the vertical direction, that is, the flow rate changes with distance more rapidly in the vertical direction.

In the lubrication regime, the approximate solution of the Navier–Stokes equations for the sphere–plane geometry leads to the following expression for the hydrodynamic force acting on the sphere approaching a wall [73]:

$$F_h = \frac{6\pi\eta r^2 v}{h}, \quad (9)$$

where v is the velocity of perpendicular approach of the sphere to the plane and h is the distance of closest approach of the sphere to the plane (see figure 3). This equation is attributed in most texts to G I Taylor (1925), even though it seems that it was never published in any of his papers (see note 38 in [24]). The Taylor equation has been experimentally verified using both macroscopic and microscopic spheres [79–81]. Note that the hydrodynamic force has the appearance of a Stokes drag force on the sphere, amplified by a factor r/h , which is $\gg 1$ in the lubrication approximation.

3.3.3. The exact solution for the sphere–flat geometry. The exact solution of the Navier–Stokes equation for a sphere–flat geometry at arbitrary distances was derived by Brenner in an analytical form [72,82]. As in the previously described cases, equations (4) and (5) were solved by Brenner under the no-slip BC. The hydrodynamic force exerted on a spherical particle of

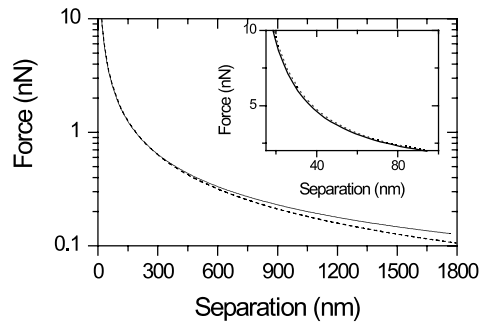


Figure 4. Semi-log plot of Taylor's solution (---) and Brenner's solution (—) for the hydrodynamic force in a sphere–flat geometry. Inset: an enlargement on a linear scale plot of the two solutions at short separations. The parameters used in the calculation of these forces were those typical for a dynamic AFM experiment: sphere radius of $10.4\ \mu\text{m}$, solution viscosity of $40\ \text{mPa s}$ and approach velocity of $2.4\ \mu\text{m s}^{-1}$.

radius r approaching a flat solid surface perpendicularly with a velocity v is given by

$$F_h = 6\pi\eta r v \lambda, \quad (10)$$

where λ is an infinite series:

$$\lambda = \frac{4}{3} \sinh \alpha \sum_{n=1}^{\infty} \frac{n(n+1)}{(2n-1)(2n+3)} \cdot \left[\frac{2 \sinh(2n+1)\alpha + (2n+1) \sinh 2\alpha}{4 \sinh^2(n+\frac{1}{2})\alpha - (2n+1)^2 \sinh^2 \alpha} - 1 \right] \quad (11)$$

and the dependence on the distance to the flat surface is explicitly expressed in the term $\alpha = \cosh^{-1}(d/r)$, where d is the distance of the centre of the sphere from the flat surface ($d = h + r$). The only approximations allowed by Brenner are those of the creeping motion, i.e. the inertial and gravitational force are ignored, as justified in the case of a small Reynolds number. The factor λ is normally greater than 1; in particular it is $\lambda \approx 1 + (9r/8h)$ when the sphere is at a distance $h \gg r$ from the wall. For h tending to infinity, λ tends to 1 and equation (10) simply reduces to Stokes formula, equation (8). As h becomes small, λ tends to r/h and therefore Taylor's approximate solution (equation (9)) and Brenner's exact solution (equations (10) and (11)) are in very good agreement at small separations. The error of the approximate solution increases with increasing surface separation. This is shown clearly in figure 4 where the Taylor solution (broken line) and the Brenner solution (solid line) for the hydrodynamic force acting on a microsphere approaching a flat surface are plotted.

The effect of a stationary wall in the path of the sphere is thus to increase the resistance experienced by the sphere beyond that which the sphere would experience in an unbounded fluid when moving at the same velocity.

3.4. Boundary conditions

The solutions presented in section 3.3 were calculated under the no-slip BC, following the derivation presented in most textbooks of continuum fluid dynamics [34, 65, 67, 68, 70, 71]. This condition is illustrated in figure 1(a), where the velocity of a moving liquid is shown to decrease linearly as the vertical distance to the lower static wall decreases, until, at contact with the solid surface, the liquid velocity is zero. The no-slip BC is commonly employed in theoretical treatments of fluid flow problems, and numerous experiments have been published that support its applicability to the flow of viscous fluids (see the historical review in section 2).

The no-slip BC is expected to break down when the important length scales traversed by the fluid molecules are nearly as short as their mean free path [67]. There are three situations where this occurs: in the flow of rarefied gases [56, 83–90], when fluids are highly confined in one or more dimensions and when new fluid–solid interfaces are being formed (liquid spreading). Employing the no-slip BC in liquid spreading situations, a stress singularity arises at the moving contact line of the liquid. The occurrence of the singularity can be alleviated by assuming a slip condition [17, 44, 91–100]. Several studies proved that even an infinitesimal degree of slip would remove the contact singularity existing in the Stokes–Einstein model of surface diffusion [43–103].

Slip at the contact line of a liquid drop moving on a substrate has been described as being due to the shear stress induced by a gradient in chemical potential in the liquid at the liquid–solid interface [104, 105]. The slip velocity can then be defined through the gradient of thermodynamic potential along the solid surface:

$$v_r = -\frac{D}{nkT} \frac{\delta\mu}{\delta x}, \quad (12)$$

where D is the surface diffusion coefficient, n is the liquid particle number density, k is the Boltzmann constant, T is the temperature and $\delta\mu/\delta x$ is the gradient of the chemical potential along the solid–liquid interface [105].

Another situation where slip must be allowed in order to avoid unrealistic behaviour is corner flow [106–108]. Here again the governing continuum equations give rise to a divergence in the velocity and stress fields if the no-slip BC is used. This is resolved by allowing for slip in a finite microscopic region with a finite wall stress.

Except for these cases, where slip was always expected even for Newtonian liquids, the main experimental evidence of the occurrence of slip has been observed in complex fluids such as long chain polymers [8, 109–123], polymer dispersions [124] and micellar solutions [125]. The mechanism for slip in polymers has been reasonably well understood [126]; it was shown that the flow of long chain polymers is fundamentally different from that of Newtonian liquids and that in polymer melts the amount of slip is proportional to the average molecular size (see references cited in [126]). On the contrary, the theories on slip in Newtonian liquids are still in development.

Recent advances in experimental techniques enable the evaluation of fluid flows with small slip lengths to be differentiated from flows with a true no-slip BC. Sections 3.5 and 3.6 present solutions of the Navier–Stokes equations that employ a partial-slip BC. This alternative BC allows for slip of the fluid at the solid boundary and therefore states that the velocity v_r of the liquid can be finite at the surface of a solid. As mentioned in section 2, the slip length b is very commonly employed to quantify the amount of slip at the solid interface, and it is related to the velocity of the liquid at the wall by equation (1) [47, 127, 128].

3.5. The slip length model

As mentioned in section 2, Navier was the first to examine the possibility that a fluid might slip at a solid surface [40]. Vinogradova [6, 127–130] has presented a derivation of the hydrodynamic force acting between two approaching spheres or cylinders in a liquid that allows for a partial slip BC. Analogous models for different geometries can be found elsewhere (see, e.g. [72, 101]). The approach by Vinogradova employs the same approximations as Taylor’s solution, i.e. the lubrication approximation ($h \ll r$) and the assumption of creeping flow. The Navier–Stokes equations are solved analytically and the resulting hydrodynamic force on a sphere has the

same form as in Taylor's formula (equation (9)), modified by a correction factor f^* :

$$F_h = \frac{6\pi\eta r^2 v}{h} f^*, \quad (13)$$

where the correction factor f^* was calculated by Vinogradova for both the symmetric and the asymmetric case. In the first case, where the drainage force between two surfaces is calculated assuming the same slip length b at both surfaces, the f^* correction is

$$f^* = \frac{h}{3b} \left[\left(1 + \frac{h}{6b} \right) \ln \left(1 + \frac{6b}{h} \right) - 1 \right]. \quad (14)$$

In the asymmetric case, where the drainage force between a hydrophobic and a hydrophilic surface is modelled, slip is assumed to occur only on the hydrophobic surface and the f^* correction is

$$f^* = \frac{1}{4} \left\{ 1 + \frac{3h}{2b} \left[\left(1 + \frac{h}{4b} \right) \ln \left(1 + \frac{4b}{h} \right) - 1 \right] \right\}, \quad (15)$$

where f^* tends to one as b tends to zero, so that the Taylor equation is recovered. Because f^* is less than, or equal to, unity, the effect of slip is that of reducing the hydrodynamic force exerted on a sphere approaching a flat surface. In other words, slip increases the rate of drainage of the fluid confined between two surfaces being driven together with a constant force. For $h \gg b$, $f^* \approx 1$, i.e. the liquid flow is unaffected by slip at surface separations much greater than the slip length. In this solution b is a constant and does not depend on shear rates or on surface separation. Since recent experimental findings have demonstrated that these conditions are not always respected [22–24, 131–135], the need for alternative models is strongly felt.

Recently Spikes and Granick [136] introduced a modified slip length model, in which a critical shear rate sets the slip regime. At shear rates above the critical shear rate, slip occurs with a finite slip length. This model is able to accommodate experimental results that appear to have a constant slip length as well as those that indicate a shear dependent slip length, as at high shear rates only the constant slip length regime is observed. Additionally, the combination of a critical shear rate and a finite slip length above this shear rate is likely to be most evident in geometries where the shear rate varies, such as the sphere–flat geometry. An alternative frictional form describing slip has been shown to yield consistent transport coefficients [137, 138].

An empirical slip parameter, α , was introduced by Ferrante *et al* to describe the relative motion of liquid molecules adjacent to a surface for quartz crystal resonators [139]. Here, α is a complex number, whose real component is equal to one for no-slip BC and approaches zero for full slip. This parameter accounts for differences in both the velocity and the phase of the liquid molecules with respect to those of the solid. Ellis and Hayward have developed a relationship that relates this parameter α to the commonly used slip length b , facilitating the comparison of results obtained with the quartz crystal microbalance (QCM) with those obtained with other techniques [140].

3.6. Flow through capillaries

Another example of commonly used geometries in experiments on slip is the flow of liquid through thin capillaries. Assuming laminar flow, the volume flow rate of a Newtonian liquid through a cylindrical capillary is given under no-slip BCs by the well-known Poiseuille formula [67, 68, 71]:

$$Q_0 = \frac{\Delta p \pi r^4}{8\eta l}, \quad (16)$$

where Δp is the pressure drop down a capillary of length l and radius r . If slip occurs, the flow rate changes to

$$Q = Q_0 \left(1 + \frac{4\eta}{kr} \right), \quad (17)$$

where k is the interface friction coefficient and it is $k = \eta/b$, where η is the viscosity and b is the slip length defined in sections 2 and 3.5. The relative change of flow rate is

$$\frac{\Delta Q}{Q_0} = \frac{4\eta}{kr} = \frac{4b}{r}. \quad (18)$$

Using these expressions, Blake [141] was able to provide a comparison, though not conclusive, between an existing slip theory and experiments performed with mercury in thin glass hydrophobized capillaries. It should be pointed out that the Poiseuille formula is often used to model the flow of blood through capillaries and arteries, and a slip BC could lead to departures from expected flow rates of up to 30% [142].

4. Experimental techniques for the investigation of BCs

The hydrodynamic BC will always appear to be one of no-slip, unless the flow is examined on a length scale comparable to the slip length. Given the small scale of the slip length measured so far (a few nanometres to a few micrometres for Newtonian liquids), no consequences of slip on the flow of liquids on a macroscopic scale are expected. Therefore, extremely accurate techniques with high spatial resolution, capable of interfacial flow measurements, are required to detect the effects of slip. This usually involves the confinement of fluid in two dimensions between surfaces, such as in capillary flows and surface force measurements, the use of surface sensitive techniques such as those employed in oscillating quartz crystal techniques or near surface imaging of flows using tracer molecules or particles. As a consequence of the nano- and bio-technology revolution, a variety of techniques are now available that are capable of probing fluid flow on these small scales and are thus suitable for the investigation of BCs.

Evidence for slip has recently been collected by a number of research groups using a range of techniques. The variety of employed methods assures that boundary slip is not an artefact of a single technique, but challenges us to explain the wide range of obtained results with an accurate physical description of events at the interface. In the following we briefly describe the most important of these techniques, with their advantages and disadvantages, and list the experimental studies that have employed them.

4.1. Techniques tracing the fluid flow near a boundary

4.1.1. Fluorescent recovery after photobleaching. The group of Léger utilized fluorescent recovery after photobleaching (FRAP) to directly observe the flow of a fluid near a solid surface (see figure 5) [21, 111, 143–145].

Fluorescent molecules are dissolved in the liquid being studied in a very low concentration and a region of the liquid is illuminated with an evanescent wave. The use of evanescent waves ensures that only the region close to the surface is probed. Flow is observed by photobleaching the fluorescent molecules with a high intensity laser beam and monitoring the recovery in fluorescence as new molecules enter the photobleached region during a shear flow. Faster recovery occurs at higher shear rates as particles are more rapidly delivered to the region of interest. Similarly, slip results in faster fluorescence recovery as the fluid motion at a given distance from the wall is faster when slip is present. Fluorescence recovery is sensitive to the

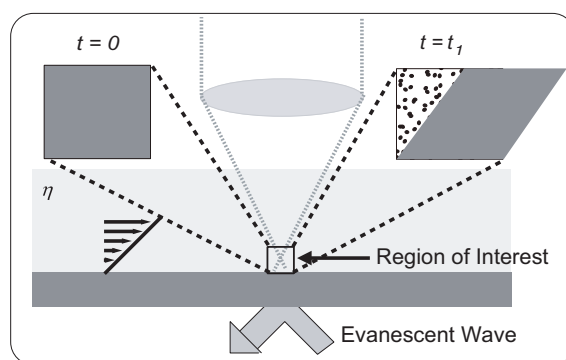


Figure 5. FRAP. Fluorescent molecules present in the liquid and in proximity to the bottom surface are excited using an evanescent wave. At time $t = 0$ the fluorescent molecules in a small region close to the surface are photobleached using a high intensity laser beam. As the liquid flows ($t = t_1$), unbleached molecules enter the region of interest, replacing the bleached molecules. The recovery of the fluorescence intensity is measured as a function of time, and information is obtained on the average velocity of the particles within the region of interest.

average of the flow velocities within the evanescent field as diffusion normal to the surface is rapid on the time scale of the experiment.

Advantages. The method is direct; the actual flow of the liquid can be followed and the geometry of the flow is simple. These two factors combined mean that the observed slip behaviour is difficult to refute.

Disadvantages. The technique does not provide an absolute measure of fluid flow: only differences in slip behaviour can be observed by comparing fluorescence recovery under different surface conditions. Therefore it is possible to quantify the slip in one system relative to another, but the slip in each system cannot be determined absolutely. The technique has a resolution of about 100 nm, determined by the decay length of the evanescent light, so it cannot distinguish small differences in slip lengths. A limited range of shear rates are accessible. Tracer molecules are required and therefore the system is ‘contaminated’, though in our opinion the tracer molecules are unlikely to have any influence on the measurements at the very low concentrations employed, as they are not preferentially adsorbed to the surface.

4.1.2. Using tracers to follow liquid flow. Several research groups have employed this approach to measure the interfacial behaviour of liquids [146–152]. Particle image velocimetry (PIV) is a well-established technique for measuring fluid velocities in macroscopic systems. Typically, the positions of tracer particles are followed and their velocity determined by statistically correlating images obtained at small time intervals using pulsed light sources. Recently, the spatial resolution of the technique has been improved in order to investigate near-surface flows on the micron scale (see figure 6).

The main barriers to such improvements are the diffraction limit of the optics and the size of the tracer particles. In order to record a measurable fluorescence signal particles must be relatively large. Meinhart *et al* have an apparatus for following the flow of a fluid near a wall with a resolution of 900 nm and a velocity vector spacing of 450 nm [151].

A similar set-up was designed by Lumma *et al*, who, using double focus fluorescence cross-correlation (FCC), were able to follow the flow profile of colloidal particles near

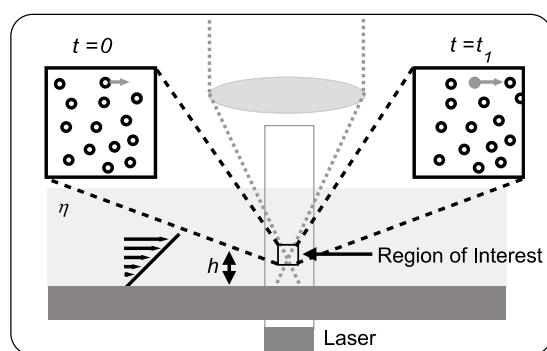


Figure 6. PIV. Fluorescent particles are immersed in a liquid and illuminated with a laser. The particles within a small volume at a given height h above the surface are imaged with a tightly focused detection system. As the liquid flows, these particles flow across the region of interest. A second image is then taken at a time t_1 after the initial image. The cross-correlation of the images is computed, which yields information on the average velocity of the particles within the region of interest.

an interface with improved resolution [150]. This technique involves measurement of the fluorescence of tracer particles as they pass through two separate lasers focused at a separation of $6\text{ }\mu\text{m}$, under the action of a flow. The flow profile is determined from the cross-correlation as well as the two autocorrelations at a number of distances from the wall. The wall position is found independently of the flow behaviour with an accuracy of about 100 nm .

Joly *et al* also used fluorescence correlation spectroscopy but they measured the spontaneous diffusion dynamics (under no shear) of colloidal tracers in a confined geometry [153]. Since the tracers dynamics is affected by the BCs at the wall, measured differences in diffusion coefficients of the tracers were converted into different slip conditions at the walls. The absence of external flow is claimed to ensure that no gas bubbles are nucleated on the hydrophobic surfaces.

Advantages. These techniques can provide an almost direct measure of fluid flow. A range of surfaces, dyes and particles can be employed in order to examine the influence of surface properties on the flow behaviour.

Disadvantages. Care must be taken in extrapolating the particle velocity profiles to the fluid velocity profile. Charge at the interfaces can lead to an electro-osmotic flow. If this flow is not suppressed by the use of high electrolyte concentrations the flow data can be adversely influenced [154]. PIV has a limited resolution as the position of the wall can only be determined to within 400 nm [151] and the particle position with even lower accuracy. Recent improvements to the technique allow the position of the wall to be determined within 30 nm [152]. Superior resolution has been reported for the FCC technique, down to a few nanometres on the measurement of the slip length [153].

Exact determination of the BC affects also the calculation of electro-osmotic flows, because the ζ -potential is defined as the electric potential at the slip plane [26, 155]. The problem of laminar electrokinetic slip flow in microchannels subject to a time-dependent pressure gradient and a time-dependent electric field has been addressed analytically [156–159]. In pressure driven flow, slip counteracts the electrical double layer effect and induces a larger flow rate. In electro-osmotic flow, slip can increase the flow rate by about 20% for a thick double layer and by more than 50 times for a thin double layer. The degree of slip depends on the solid material,

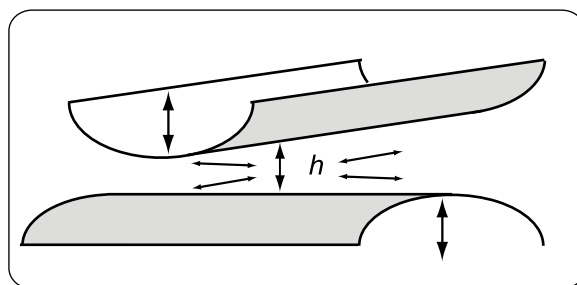


Figure 7. SFA. Atomically smooth mica surfaces are attached to cylindrical lenses which are arranged perpendicularly and immersed in a liquid. The distance between the surfaces h is measured using fringes of equal chromatic order (FECO) or a capacitance gauge or bimorph. The drainage force arising from the normal approach or oscillatory motion is measured using the deflection of a spring on which one of the surfaces is mounted.

electrolyte concentration and pH. Similar conclusions were reached for electro-osmotic flow through hydrated clays [4].

4.2. Techniques based on force or displacement measurement

Here we group the techniques of AFM, SFA and a number of other similar apparatuses. Craig has published a review of the development of force measurement techniques up to the 1990s [160]. These instruments are used to measure either the hydrodynamic drainage force between two approaching surfaces or the displacement of one surface in response to the oscillation of the other surface.

4.2.1. Surface forces apparatus. Several groups have used the SFA to measure the drainage force between two approaching surfaces [23–25, 133–135, 161–164]. The SFA was originally designed to measure surface forces between two half cylindrical mica surfaces (see figure 7) [165–167].

The separation between the atomically smooth mica surfaces can be measured by the fringes of equal chromatic order technique (FECO) with sub-nanometre resolution. Typically one surface is mounted rigidly on a piezoelectric transducer and the other on a calibrated spring. In early measurements of drainage force the surface mounted on the transducer was driven a known distance for a set period and then the drive was stopped. The surface separation was monitored as the fluid of interest drained out from between the two surfaces [73, 161, 168–172]. In later models oscillatory measurements were performed [23, 25, 173, 174], whereby the piezo mounted surface was driven normal to the surfaces with a known amplitude and frequency and the gap between the surfaces was monitored at a range of surface separations. A surface can also be oscillated parallel to the other to perform measurements in ‘friction’ mode [175, 176]. Rather than monitor the surface separation using FECO, it is possible to monitor the response of one or both surfaces either using highly precise capacitance gauges or piezoelectric bimorphs [177]. Linear variable displacement transducers can also be employed. Other techniques using macroscopic surfaces similarly employ capacitance gauges [178].

Advantages. This well-established technique features very high resolution in force and displacement and a range of accessible shear rates. Atomically smooth surfaces are routinely used. When FECO is employed the interaction area can be inspected for the presence of

contaminants such as micron-sized particles. Oscillatory measurements readily yield a slip length and drainage measurements can be analysed using the Vinogradova models of slip (see section 3.5).

Disadvantages. A limited range of materials can be employed. The geometry of interaction results in a complex flow field, which complicates any analysis of shear rate effects. Some of the studies of fluid drainage are likely to be influenced by the presence of contaminating platinum particles on the mica surface [179, 180]. These particles arise when the mica surfaces are cut using hot platinum wires and have been shown to have considerable effects on frictional measurements performed with the SFA [181, 182]. The presence of these particles can be eliminated by using appropriate experimental protocols [182–184].

4.2.2. Atomic force microscopy. Atomic force microscopy (AFM) (or similar custom-built instruments) has been used in a similar manner to SFA to measure drainage forces between two approaching surfaces [22, 26, 74, 75, 130–132, 185–187]. Surface forces can be measured by placing a silica sphere on the tip of the AFM cantilever (see figure 3) [188, 189]. The substrate is driven towards the sphere and the deflection of the cantilever is monitored using a light lever technique, which results in a very accurate measurement of surface forces. The surface separation is determined from the calibrated piezo displacement using a zero of separation determined from the force data [189]. Measurement of equilibrium surface forces are performed quasi-statically, using low piezo ramp rates. Employing high piezo ramp rates, hydrodynamic drainage forces can be measured over a large range of shear rates [22, 26]. As with the SFA technique, it is possible to perform oscillatory measurements using the AFM technique [132].

Advantages. The technique is very accurate and can be used with a wide range of substrates, including moderately rough surfaces [74]. The error in the determination of the zero of separation is normally around 1–2 nm (an order of magnitude less accurate than SFA, but still very good compared with other techniques). The drainage data can be analysed within the Vinogradova framework to arrive at a slip length in the same manner as SFA experiments. When oscillatory measurements are performed, the slip length can be determined readily using the approach of Israelachvili [173], although the gap distance is not measured directly. A larger range of shear rates than accessible with the SFA can be employed.

Disadvantages. These measurements are complicated by a number of factors. The approach rate of the surfaces is not constant, even if the piezo drive rate is constant, and has to be calculated from the piezo drive rate and the rate of cantilever deflection. The deflection increases rapidly near the surface due to the increasing hydrodynamic force; therefore the approach rate in this region drops considerably. When rough surfaces are employed the accuracy in the determination of the zero of surface separation is not as good [74]. Oscillatory measurements are at this point in time challenging due to difficulties in fully characterizing the instrumental response [132]. Inertial effects have been shown to increase when the time scale of the approach motion becomes similar to the inverse resonance frequency of the cantilever [78]. Also, the cantilever contributes to the total hydrodynamic force measured with an AFM [75]. However, Neto *et al* [131] calculated that in most cases this contribution requires a very small correction on the calculated slip length. Finally, the curvature of the employed surfaces might influence the measured slip length (this issue is discussed in section 5.5.2).

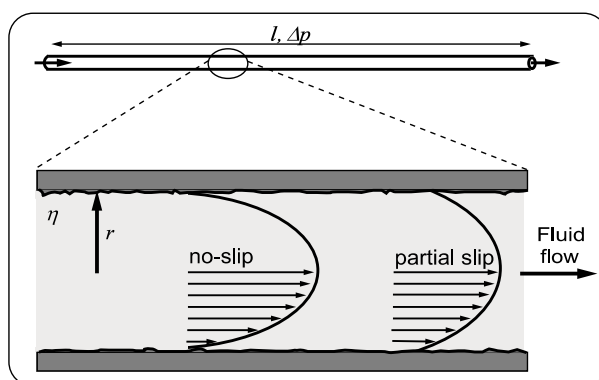


Figure 8. Capillary technique. A liquid is driven through a capillary of radius r by a pressure difference Δp along its length l . The flow rate of the liquid is measured and compared with predicted values (equations (16) and (18)).

4.3. Capillary techniques

Numerous experiments on the slip BC were conducted using thin capillaries [16–18,190–205]. Accurate measurements of liquid transport through a capillary can be used to determine the BCs (see figure 8).

A slip BC increases the fluid flow compared with a no-slip BC, in a measure that is proportional to the slip length (see equations (16) and (18)). The effect of different factors, such as pressure, surface roughness and geometry, on the flow of simple liquids in microchannels has been investigated, but a model that is able to explain all experiments has not yet been proposed [206]. Rapid improvement in flow rate detection methods [205], combined with improved methods for the fabrication and characterization of capillaries, may result in this technique being extremely useful. Track-etched membranes, which are produced by irradiating a material with charged fission fragments followed by chemical etching, may prove very useful for flow studies [207]. The dislocations caused by the irradiation provide pathways that are susceptible to etching and thereby can be used to determine the location of the long axis of the extremely narrow capillary. The width of the capillaries is controlled by the time of etching as the lateral growth is much slower than the initial longitudinal etching. Unfortunately the analysis of the flow is complicated by the fact that these capillaries are not cylindrical.

Advantages. The technique is very simple, in its basic form does not require elaborate equipment and readily yields a slip length.

Disadvantages. The greatest source of error in measurements of flow through microchannels comes from the measurement of the inner diameter of the channels. The fabrication of channels that are atomically smooth, of constant known width and very clean remains problematic. Also, in the case where the surfaces are modified by adsorption of surfactants or self-assembled monolayers (SAM), the coverage and wettability inside the channels cannot be tested independently. Future developments of the technique are likely to result in a very complex instrumental arrangement. Electro-osmotic flow may complicate the analysis of some systems: indeed Grimes and Liapis [208] proposed a slip model for electro-osmotic flow in open tube capillary electrochromatography systems and found that the occurrence of slip would significantly alter the electro-osmotic flow.

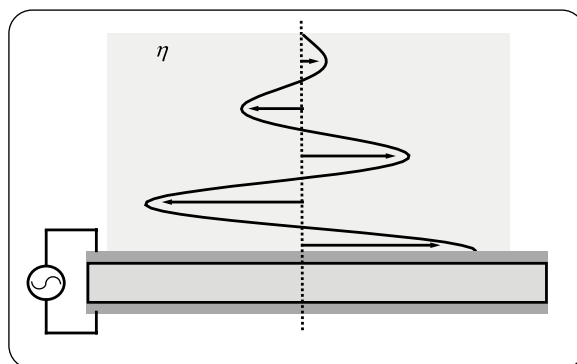


Figure 9. QCM. The liquid of interest is in contact with one side of a piezoelectric quartz crystal, which is being driven at its resonance frequency. The resonance frequency and damping of the quartz crystal depend on the acoustic impedance of the material near the surface and coupling at the interface. Analysis of the change in the damping and resonance frequency of the crystal for numerous overtones is used to infer the slip condition at the boundary [29,211].

4.4. Quartz crystal resonators

The BCs of a liquid flowing past a solid surface have been studied using resonating quartz crystals [139, 140, 209–214]. A quartz crystal driven at or near its resonance frequency can be used to probe fluid properties and the adsorption of material to the surface of the crystal. In this mode it is often called a QCM and provides a very sensitive measure of the amount and viscoelastic properties of material adsorbed at the interface (see figure 9). The presence of a liquid in contact with the surface of the crystal dampens its oscillation and changes the resonant frequency, by changing the acoustic impedance of the device. Since this effect depends strongly on the degree of coupling at the surface–liquid interface, QCM can be used as a sensitive technique to probe boundary slip [29, 139, 211]. QCM is also employed to investigate the slip behaviour of monolayers of material [215], but this aspect is outside the interests of this review. Dialogue between the QCM community and other researchers interested in slip has been scarce so far; however, the situation might improve thanks to recent attempts, such as that of Ellis and Hayward who have related the slip measured in QCM systems to the commonly used slip length parameter, b [140].

Advantages. Typically, crystals oscillate at 5 MHz or higher with nanometre amplitude and therefore high shear rates are accessible. Results are obtained easily and with great precision. A range of substrates can be used and the influence of a wide range of adsorbates can be studied. The technique enables other properties of the film to be probed simultaneously.

Disadvantages. It is unclear how the finite surface roughness of the layer of gold that coats the crystals may affect the response of the crystal and the interpretation of slip. Relating the crystal response to the slip behaviour and a slip length is non-trivial and is necessarily model-dependent though theoretical developments are continuing apace [29, 211].

4.5. Other techniques

Other methods have been used to study liquid/solid BCs, including studying the flow of thin films on spinning disks [216] and on rotating cylinders [217], droplets moving down an inclined surface [218], particle sedimentation rates [7, 219] and excitation of surface waves [220].

5. Factors affecting slip

It has long been accepted that the nature of the liquid–solid interface, and in particular surface wettability and roughness, has a strong effect on the slip behaviour. More recently, attention has been drawn to the influence that the shear rate and the presence of gaseous layers at the interface might have on slip. All these factors are strongly connected and their effects cannot be easily decoupled:

- (i) The macroscopic contact angle, which is commonly used as a measure of wettability, depends not only on the chemical nature of the surface but also on its roughness. Surface roughness magnifies the intrinsic wetting properties of a surface, enhancing both the solvophilicity or solvophobicity of a surface [221]. The main effect of surface roughness on contact angle is described by the Wenzel equation [222], which states that the apparent contact angle ϑ^* of a drop of liquid on a rough surface is related to Young's intrinsic angle ϑ_e on a smooth surface of the same material by

$$\cos \vartheta^* = r \cdot \cos \vartheta_e. \quad (19)$$

The parameter r is defined as the ratio of the real surface area to the apparent (projected) surface area and quantifies the surface roughness. Because $r > 1$, Wenzel's equation states that roughness improves wettability of a wetted or partially wetted surface, that is:

$$\vartheta^* < \vartheta_e \quad \text{for } \vartheta_e < 90^\circ. \quad (20)$$

On the other hand, roughness reduces wettability for a non-wetted one:

$$\vartheta^* > \vartheta_e \quad \text{for } \vartheta_e > 90^\circ. \quad (21)$$

- (ii) A chemically heterogeneous surface could lead to changes in the average slip behaviour because of the different BCs valid in different domains.
- (iii) The stability of gas layers and nanobubbles is higher on high contact angle surfaces and rough surfaces. Indeed superhydrophobicity is attributed to the ability of very rough surfaces to trap a layer of gas beneath the non-wetting fluid [223].
- (iv) Local shear rates for a fluid flowing over a surface strongly depend on surface roughness, because in a confined geometry an asperity on the surface might reduce the gap where the flow occurs, therefore increasing locally the shear rate; the opposite is true for a crevice.
- (v) The presence of adsorbed species, such as surfactants, polymers or nanobubbles could vary both the surface wettability and roughness of the surface; these adsorbates could also displace the slip plane inside the liquid at a distance from the true solid interface.

All the possible effects of surface modification need to be considered jointly, and failure to do so has probably resulted in the lack of agreement between published results. It is now becoming more clear that any study of liquid slip should build on a thorough characterization of the interface both from the chemical and from the physical point of view. Bearing in mind that all these parameters are strongly coupled, in the following sections we will review the studies that investigate these effects separately.

5.1. Surface wettability

From the beginning of the debate on BCs, it has been assumed that liquids are more likely to slip on poorly wetted surfaces. A high contact angle is indicative of a weak interaction between liquid and solid, that can be more easily overcome, causing the fluid molecules to slide across the solid. Indeed initially it was mainly experiments performed with hydrophobized surfaces or metal surfaces (easily contaminated to become hydrophobic) that gave support to

a slip BC. In 1952 Tolstoi [224] attempted to justify slip in terms of the underlying molecular-kinetic mechanism and proposed that the mobility of the liquid molecules adjacent to a solid surface depends on the equilibrium contact angle of the liquid on the solid. This approach was extended by Blake, who reaffirmed the concept that the degree of boundary slip is related to the contact angle of the fluid at the interface [141]. This approach was later applied to quartz crystal resonators [213]. Many molecular dynamics (MD) simulations see an increase in slip when the strength of the molecular potential between the fluid and wall molecules is reduced [225, 226]. However, the slip lengths estimated by simulations are normally much smaller than those measured in physical experiments.

Several of the first reports on liquid slip and a few of the recent ones suggest that slip only occurs on non-wetted surfaces [18, 20, 25, 75, 146, 162]. For example, Baudry *et al* [25] found that the BC changed from one of no-slip on a cobalt surface to one of slip when the surface was hydrophobized with a gold–thiol treatment ($\theta = 101^\circ$, $b = 40$ nm; note however that the employed liquid, glycerol, is hygroscopic, therefore subject to changes in viscosity that would affect the results). Choi *et al* [199] examined the flow through microchannels and found an increase in slip length for hydrophobic channels. Zhu and Granick [135] found that addition of surfactant to mica-alkane systems resulted in an increase in contact angle and in slip, though the contact angle was rather low ($<40^\circ$) in these systems. Cottin-Bizonne *et al* [162] found a moderate degree of slip (20 nm) for hydrophobic surfaces and a no-slip BC for hydrophilic surfaces.

Even within the same techniques, results appear inconclusive. Pit *et al* found a higher degree of slip on a sapphire surface after it was methylated [21] or modified by adsorption of steric acid [143], but the degree of slip was reduced when the surface was fluorinated.

Several reports of very large slip lengths using hydrophobic surfaces can be attributed to the presence of gaseous material at the interface [159] and should be treated separately from evaluations of slip in other systems. For example, Tretheway and Meinhard attributed the measured apparent slip lengths of $1\ \mu\text{m}$ in hydrophobic microchannels [146] to the presence of gas at the interface. A report of large slip lengths between hydrophobized pyrex surfaces [162] was later also attributed to the presence of nanobubbles [227].

At the opposite side of the spectrum, some papers show very clearly the occurrence of slip also at the interface with surfaces which are partially or totally wetted by the employed liquids [21, 22, 26, 131, 135, 143, 164]. Bonaccorso *et al* demonstrated the presence of slip on a completely wetted silica surface [26, 74]. Henry *et al* [132] used different concentrations of physisorbed surfactants to modify surfaces so that the contact angle first increased and then decreased. They found that the slip length and contact angle were not correlated, and that the shear dependence of the slip length was changed by the amount of surfactant at the surface. The elegant study by Cho *et al* [186] could explain some reported discrepancies: they found that the degree of slip is related to the contact angle only for n-alkanes and not for other classes of liquids, such as non-polar ring shaped molecules and polar liquids.

In view of these contrasting results, the influence of wetting properties on slip certainly needs to be further investigated. Currently the evidence would suggest that the level of hydrophobicity is not a primary factor in determining slip [27], despite the fact that many researchers have seen an increase in slip after modifying the wettability of a surface.

5.2. Surface roughness

It is intuitive that boundary slip should also be linked to surface roughness, particularly as corner flow is associated with slip, but so far discrepancies between published experiments are not resolved. Very few experimental studies have concentrated on investigating the influence

of roughness on boundary slip. These experiments are challenging for a number of reasons, including the following:

- (i) it is difficult to produce suitable surfaces of controlled roughness.
- (ii) most efforts to alter the surface roughness result in additional undesired changes to the interface properties.
- (iii) the interpretation of the results is complicated by the uncertainty in the wall position that is associated with roughness.
- (iv) an appropriate theoretical description of realistic surface roughness is yet to be found.

Ideally, studies of liquid slip versus roughness should employ surfaces prepared with etching and photolithography processes, which do not alter the wettability or the elasticity of the surfaces [203, 228]. Most indications that molecular-scale roughness might suppress slip [21, 133, 145, 229] or its shear-dependence [132] were obtained by partial monolayer coverage or adsorption of polymers, which probably affect the BC in a number of other ways. Bonaccorso *et al* were the first to study the influence of roughness on a larger scale using chemically etched silica surfaces and found that slip increased with roughness [74]. The observed dependence of slip on roughness was not changed by changing the position of the wall within reasonable limits.

Some results from the QCM community are also inconclusive: McHale and Newton [212] found that a rough crystal surface suppresses slip of Newtonian wetting liquids, while Du *et al* [211] found that in a similar situation slip is present, and that it increases with increasing shear rate and with increasing viscosity of the liquid. In the engineering community, the effect of surface roughness on the flow of simple liquids in microchannels has been thoroughly investigated, but a model valid to explain all experiments has not been proposed (see [230, 231] and references therein).

Theoretical calculations have provided conflicting results also, some predicting that roughness would increase the degree of slip [102, 232–235], some that it would decrease it [236–241] and some that the relationship between roughness and slip is more complex and depends on the scale of roughness [242]. In the first category, Hocking [102] found that the irregularities on the surface give rise to apparent slip at the moving contact line of a liquid, because the displaced fluid is trapped in the crevices of the surface, therefore facilitating slip of the moving liquid. In the second category, Richardson [236] used a very simple model of a rough wall to conclude that the energy lost through viscous dissipation as a fluid passes over and around surface irregularities is sufficient to ensure that it is effectively brought to rest. Bocquet and Barrat [243, 244], Gao *et al* [245] and Miksis and Davis [246] found that even a very small wall corrugation suppresses slip. In the third category, Galea and Attard [242] and Jabbarzadeh *et al* [238] found that roughness influences slip in a complex manner: the BC is one of no-slip for commensurate structures only (i.e. when the liquid molecules are of comparable size to the surface structures) and tends to one of slip for rougher and smoother, but non-commensurate, surfaces.

Clearly the extensive theoretical work indicates that the magnitude and the length scale of roughness are important, but there is no general agreement as to what influence roughness has. We note that efforts have concentrated on near-molecular scale roughness because comparatively large scale roughness (several nanometres) cannot easily be investigated using MD techniques due to the computational expense of using large systems. Alternative techniques such as Lattice–Boltzmann methods [247] may prove to be useful in exploring larger scale roughness that is more similar to most real systems.

5.3. Shear rate

The effect of shear rate on liquid slippage is still a controversial element in the debate on BCs. In the original Navier model [40], and in the model by Vinogradova [127, 129], the slip length does not depend explicitly on shear rate. Some experiments performed recently agree with this constant slip length model [21, 26, 75, 143, 152, 161, 162]. However, several measurements are in contrast with this idea. Experiments performed with the SFA agree with others performed with the AFM in showing a shear-dependent slip [22–24, 74, 131–135, 229]. Another category of experiments found explicit evidence of shear-dependent slip in hydrophobic microchannels and less clear evidences (due to the resolution limits of the technique) in hydrophilic microchannels [199].

The pioneering simulation by Thompson and Troian [248] found that the Navier model is only correct at low shear rates; at high shear rates the slip length starts to increase abruptly with increasing shear rate. The critical shear rate at which the slip length diverges decreases with decreasing surface energy corrugation, and so is system-dependent (similar results were also obtained for confined polymers under shear [120]). Importantly, the BC becomes non-linear even though the liquid remains Newtonian. It should be pointed out that the critical shear rate was found to be very high for simple liquids, some 10^{11} s^{-1} for water, such that it can only be achieved experimentally in very small devices at very high speeds. The MD simulations performed by Stevens *et al* [249] at very high shear rates (currently not accessible experimentally) and under very high confinement of the fluid showed that, even though the amount of slip decreases as the strength of the wall–fluid interaction increases, a residual amount of slip is always observed.

The shear dependence of slip could provide a conciliatory explanation as to why several earlier experiments on confined liquids have not observed evidence for boundary slip [73, 173, 250, 251]. Under the low approach rates previously employed we expect the slip length to be effectively zero and a no-slip BC to be applicable. Only a few attempts have been made so far to theoretically explain the observed shear-dependence [136, 159, 252]. A promising explanation is that there is a threshold value of the shear rate above which slip is detectable [10, 23]. Recently Priezjev and Troian proposed that the dynamic behaviour of the slip length should be the same for both simple and polymeric fluids [253].

Lauga and Brenner claim that the effects due to gas films at the surface are dynamic, and the shear-dependence of slip is induced only in dynamic drainage measurements [159]. This model would explain the observed shear dependence of slip in some experiments. They describe a gas ‘leaking-mattress’ effect, where surface bubbles grow in response to an oscillatory interaction by rectified diffusion.

5.4. Nanobubbles or gaseous films

Several authors believe that the observed increase in flow rates in Newtonian liquids is to be ascribed to the presence of small amounts of gas trapped or pinned on rough and/or hydrophobic surfaces [127, 159, 252, 254, 255]. The ‘effective’ slip length then provides a means to average the flow over a heterogeneous surface, where regions of perfect slip (liquid flowing over gas, $b = \infty$) exist along with an otherwise no-slip ($b = 0$) surface [256]. The presence of adsorbed gas on hydrophobic solid surfaces is nowadays also considered to be the cause for most observed long-range hydrophobic attractions [257–262]. It is interesting to note that the debate over the long-range hydrophobic attraction has many parallels to that on hydrodynamic slip: firstly, it has only become an important issue since the development of highly sensitive

techniques; secondly, there is great variation in published results and thirdly the behaviour is clearly sensitive to the surface preparation.

Early experiments using capillaries often suggested the presence of a gaseous film at the interface, particularly for hydrophobic surfaces. This led Ruckenstein and Rajora [98] and Vinogradova and coworkers [263] to theoretically investigate the flow of liquids in capillaries with inhomogeneously hydrophobized walls. Ruckenstein and Rajora concluded that slip occurs over a ‘gap’, not directly at the solid–liquid interface. Vinogradova *et al* attributed the origin of slip to the decrease in viscosity in a very thin layer adjacent to a hydrophobic wall [263]. In a later paper Andrienko *et al* [254] attributed slip to a prewetting transition, i.e. the formation of a film of vapour or phase-separated ‘lubricant’ with lower viscosity between the fluid and the solid wall.

The stability of nanobubbles, even on hydrophobic surfaces, is not easily explained. Spherical gas microbubbles in liquid are unstable because the pressure inside the bubble is much higher than in the surrounding solution and this should increase the gas solubility, resulting in rapid dissolution [260, 264, 265]. However, there is considerable evidence from direct AFM imaging [257, 266, 267] and other techniques [262, 268–271] that nanobubbles can remain stable on solvophobic surfaces. Furthermore, there is growing evidence that nanobubbles are also present at completely wetting interfaces, such as water on mica [266, 272, 273], which raises the prospect that all surfaces submerged in liquid may under some conditions bear nanobubbles.

With the exception of the report by Attard and Tyrrell [261, 274], nanobubbles have been reported as isolated bubbles on the surface. We note here that two of us have disputed the interpretation of the images presented by Attard and Tyrrell, claiming that the structures that they observed are liquid-like polymer films that result from the treatment of the surface with dichlorodimethylsilane [275]. Indeed tapping mode AFM images of nanobubbles obtained in our lab on a similar surface have shown that the nanobubbles have smooth contact lines and appear similar to other reported nanobubbles [276], unlike the structures reported by Attard and Tyrrell. Regardless, in most cases where nanobubbles have been observed the surface coverage is low. This finding does not support the recent model proposed by Lauga and Brenner, who found that a nearly complete coverage of the surface by nanobubbles would be required to explain the observed slip effects [159].

If nanobubbles or a gas layer are present at the interface, it is often assumed that an infinite slip BC will apply. However, it is well known that the air–water interface rapidly accumulates contaminants [277], which could indeed increase the stability of the interface by forming a ‘skin’ at the gas–liquid interface. This effect is less noticeable in solvents other than water. The presence of such skin may act as a barrier to dissolution, provide some degree of mechanical strength, influence the surface forces and alter the BC from one of infinite slip to partial slip (and possibly, though it is unlikely, of no-slip). Further, due to the presence of a skin consisting of contaminant materials the slip condition at the surface of a nanobubble is likely to vary from experiment to experiment and depend on such factors as the total surface area of the nanobubbles, the solvent and the cleaning procedure. It appears that gas layers and nanobubbles might play an important role in measurements of boundary slip, but examining their influence is likely to provide a major challenge to experimentalists, due to difficulties in controlling their production, density, size and surface properties.

Zhang *et al* have recently proposed a new approach to the control of nanobubble formation that involves the manipulation of the solubility of dissolved gases [273]. The authors showed that a great number of nanobubbles can be formed in water on graphite surfaces (and a smaller number on mica surfaces) if the substrate has been previously exposed to ethanol. Figure 10 shows nanobubbles imaged by tapping mode AFM on a mica surface after ethanol/water

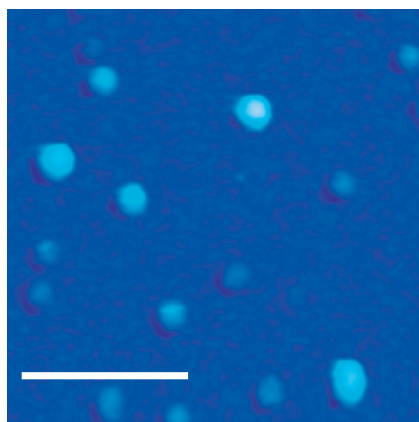


Figure 10. Tapping mode AFM image of nanobubbles captured in water on a mica substrate. The scale bar is 400 nm and the vertical scale is 35 nm. Image courtesy of Xue Hua Zhang, Department of Applied Mathematics, Australian National University.

exchange. The nanobubbles are approximately 50–100 nm in diameter and 5–20 nm high. Using variations in temperature and solvent the number density of nanobubbles can be influenced. Importantly the use of solvents that have been outgassed results in the absence of nanobubbles, strongly suggesting that outgassing be incorporated into future studies of boundary slip.

The work of Zhang *et al* [273] sheds new light on experimental procedures previously employed to prepare clean surfaces. Two of us recently prepared SAM of alkanethiols on gold *in situ* by filling the AFM liquid cell with an ethanol–thiol solution, then flushing with pure ethanol and then with pure water [22]. This procedure prevented the surfaces from being exposed to air, so as to minimize both contamination and trapping of air at the surfaces. Equilibrium force curves were performed to assure that no long-range attraction could be measured between the prepared surfaces. (Baudry *et al* also took such precautions to ensure that nanobubbles were not influencing their measurements of boundary slip [25].) However, in light of the results of Zhang *et al*, we must now acknowledge the possibility that the surfaces employed by us may have presented some nanobubbles, and that this may have influenced the measurements. Many published studies may indeed be affected by nanobubbles or entrained air at the interface [146, 227], regardless of the surface wettability and roughness, though clearly some conditions will favour the presence of nanobubbles. For example, Cottin-Bizonne *et al* [227] have recently reported that they believe their earlier work [162] was affected by gas at the interface. Many more targeted experiments are required. For the present it seems likely that studies reporting slip lengths exceeding about 100 nm might be affected by the presence of nanobubbles.

A separate issue is the possible role that dissolved gases may play in inducing boundary slip. Lichter *et al* [278] argue that slip occurs by two mechanisms: localized defect propagation and concurrent slip of large domains. Dissolved gases, ubiquitous in most liquids, must provide interfacial ‘defects’ and therefore could increase slip.

5.5. Other factors affecting slip

5.5.1. Adsorption of species from solution. There is contradiction among the published studies on the effect of the adsorption of surfactant molecules to solid substrates on the degree

of slip. Evidence was provided that when surfactants are physisorbed on a solid surface, the degree of slip becomes independent of the degree of surface coverage and hence on surface wettability [132]. Slip was also shown to be simply enhanced when surfactants are adsorbed on the surface [135]. However, these studies were conducted in aqueous and organic solvents, respectively, and therefore should only be compared with caution. Furthermore, surface modification by adsorption of surfactants presents a few complications: the surfactants often physisorb (e.g. on silica and mica surfaces) in the form of patches and aggregates [279]. This means that surfactant adsorption modifies not only the chemistry but also the molecular scale roughness and the local compliance of the surfaces. Surfactant molecules at the interface are mobile and may influence the local flow behaviour both by dissipating energy and by creating surface flows in response to induced gradients in surface excess [280].

5.5.2. Curvature. A few publications [281–283] have highlighted that the curvature of the solid boundary may also affect dramatically the liquid slip behaviour. Quantifying the relevance of the curvature effect could resolve discrepancies in the magnitude of the slip length measured with the AFM and the SFA. It is appropriate to note that at least one of these discrepancies can be explained with errors in the data analysis [284]: the first SFA study of Zhu and Granick reported slip lengths of up to $2\ \mu\text{m}$ [23], much larger than the 20 nm observed by Craig *et al* in similar wetting conditions [22]. This error has probably led to an overestimation of the fitted slip length, and was amended in a later paper [133], where slip lengths of a few tens of nanometres were reported.

5.5.3. Polarity of the solvent. Cho *et al* [186] studied the slip behaviour of several liquids and found that for highly polar molecules the slip length decreases with increasing dipole moment. They proposed that this is due to the formation of a ‘surface lattice structure’ in the liquid between the two approaching surfaces due to dipole–dipole interactions. These experiments raise many interesting questions; however, they do not explain differences in slip behaviour among non-polar solvents. The arguments relating solvent–solid commensurability with slip in MD simulations are weakened by these experiments [285–287].

5.5.4. Viscosity. Apparent slip could arise in a thin stratum of liquid of lower viscosity formed next to the wall of a smooth solid surface or in regions of higher shear next to the peaks and ridges of a rough solid [94]. In particular, this is thought to occur for highly hydrophobic surfaces [127]. However, experiments [171, 173–291] and simulations [292] have shown that the viscosity of simple liquids is in close agreement with the bulk value down to very small separations (about 10 molecular diameters).

Craig *et al* found that the slip length also increases with increasing viscosity of the liquid [22]. Du *et al* confirmed this finding using quartz crystal resonators and suggested that the interfacial friction coefficient k ($k = \eta/b$), rather than the slip length b , may be a more appropriate measure of slip.

6. A mechanism for liquid slip

Many questions regarding slip in Newtonian liquids are still unanswered, including the most basic of them: how does slip actually occur? Can we draw a simple cartoon that summarizes the basic physics of the problem and makes sense of all the experimental evidence so far collected? Unfortunately we are not in a position to answer this question at present, and to the best of our knowledge nobody can. We can attempt to summarize the few models that have

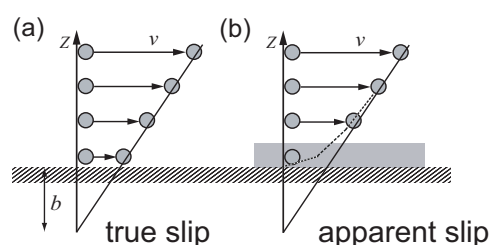


Figure 11. (a) Schematic representation of the mechanism of true slip and (b) apparent slip. The length of the arrows is proportional to the velocity of the liquid and the circles indicate molecules in the liquid before and after flow has occurred at the liquid/solid interface. The shaded region indicates a thin layer of liquid or gas attached to the solid wall.

been proposed so far, but no model is universally believed to be correct. Two simple pictures can be drawn that could explain the observed reduction of the frictional force between a liquid and a solid:

- (i) true slip at a molecular level, where liquid molecules are effectively sliding on the solid surface (see figure 11(a)).
- (ii) apparent slip, where the slip occurs not at the solid/fluid interface but at the fluid/fluid interface (see figure 11(b)). This happens if a thin layer of liquid molecules is tightly bound to the solid surface, but the velocity gradient close to the solid is so high that the molecules beyond this layer appear to slide on the surface. The thickness of this layer can be as small as a single monolayer, or larger, in which case this layer is known as a stagnant layer (as originally proposed by Girard, see figure 1(b)). We also include in this picture the situation where slip occurs on top of a static layer of a different phase, for example, of gas or a less dense liquid.

At present, to the best of our knowledge, no experimental technique is able to distinguish between the two pictures and to directly measure the motion of molecular layers of liquids close to a surface. All experimental results are based on indirect measurements, and they point to the fact that viscous drag is lower than would be expected based on predictions of continuum theories. Therefore any model for slip that might be proposed needs to account for this reduction of drag. For the same reason, all models which have appeared up to now are only phenomenological and cannot explain the MD underlying the observed phenomenon. In the following paragraph we list the most accredited mechanisms by which true slip could occur:

- if the viscous friction between liquid molecules at the interface is stronger than between molecules of the liquid and molecules of the solid, then the molecules can slide on the surface. This is true for lyophobic surfaces [20], but might also hold, under certain circumstances, for lyophilic surfaces [143].
- if the dimensions of the liquid molecules are of comparable size to the rugosity of the solid surface, the molecules are trapped in the pits on the surface. If their size is much smaller or much larger, they can slide on the surface. This is true for two solids sliding along each other [293], but simulations show that it might also hold for liquids on solids [242, 294].

If the picture of apparent slip is correct, there are a few ways by which a boundary layer with a modified viscosity and/or mobility could be created:

- at room temperature and pressure, there is always some residual gas dissolved in a liquid. Critical levels of shear might induce cavitation in a liquid and the generated gas bubbles

might adhere to the solid surface, forming a layer at the interface onto which the liquid can slip [23, 159, 252, 295].

- at a critical shear rate, a microscopic surface roughness or corrugation can favour the generation of a layer of turbulent flow at the interface, and thus modify the viscosity of this layer with respect to the bulk, even if the overall flow is laminar [296]. An example from nature is the so-called shark-skin effect [297].
- ordering of liquid molecules at solid surfaces reduces the friction between neighbouring layers and therefore facilitates the sliding of one layer of molecules past the other. The ordering is strongest the closer the molecules are to the solid surface [185, 298].

7. Outlook

Ultimately, this review highlights the strong need for an accurate physical description of fluid flow near an interface with molecular resolution. However, current techniques do not appear to be capable of yielding such results, and therefore we depend on indirect techniques to infer fluid flow near interfaces. At the same time, there is a strong need for theoretical and simulation studies in the area that, overcoming the current phenomenological description of slip, might provide a basis for understanding the experimental results.

A major challenge for experimentalists is to decouple the effects of roughness, wettability and nanobubbles in investigations of the fluid–solid BC. This is difficult because of the paucity of reliable data on the production and stability of nanobubbles and on the influence of nanobubbles on the BC. The removal of dissolved gases from solvents can be used to minimize the production of nanobubbles and presently appears to be the most effective approach to eliminate the influence of nanobubbles. A controlled study of the influence of nanobubbles on boundary slip remains problematic as the production of nanobubbles is still irreproducible. The nanobubble–liquid interface is likely to be contaminated and highly variable and will in turn influence the nanobubble–solvent BC.

The production of rough surfaces with controlled morphology suitable for studies of the BC remains a significant challenge. Ideally, experiments should employ surfaces that can also be studied via theoretical and computational approaches. Additionally, rough surfaces implicitly complicate measurements of boundary slip, as the zero of separation between rough surfaces is poorly defined.

In order to compare results of different studies on roughness, it would be desirable to find an appropriate description of a rough surface. Measures such as the root mean square roughness or the peak to peak amplitude of the roughness are not very useful. An analysis based on the local curvature of the surface is more likely to correctly describe fluid flow, as slip is expected around features with a high curvature (as in corner flow [106–108]). Theoretical studies often employ surfaces that have regular features, such as sinusoidal corrugations, that can be described with a wavelength and amplitude. The challenge for theoreticians is to mimic real surfaces that have roughness that is neither randomly generated nor periodic (see figure 12). A few promising approaches have been made in the right direction, modelling the roughness as a layer of porous material [230] and as a series of randomly distributed bosses [240]. Many more careful studies will be required before the issue of surface roughness can be resolved.

Several simulation studies have addressed the issue of liquid–solid ordering and commensurability [285–287], but only a few experimental attempts have been made in this direction [186, 298]. More experiments are definitely required: atomically smooth crystalline surfaces such as mica and sapphire could be used as substrates and molecules of different shapes and/or polarity could be employed to observe the liquid ordering or to alter the liquid–solid epitaxial matching.

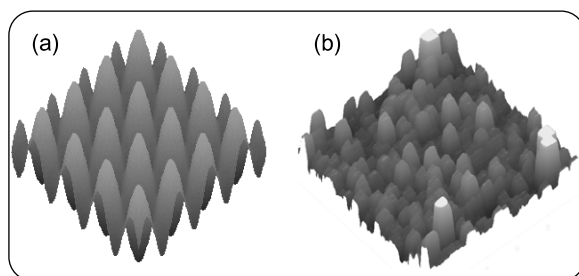


Figure 12. (a) An ideal rough surface. (b) A real rough surface (etched silicon wafer) imaged by AFM. The roughness of a surface is most commonly characterized using measures such as the peak to peak amplitude and the root mean square roughness. In the ideal case (a), the surface roughness is a constant across the whole surface. In the real case (b), the roughness depends on the region of the surface that is observed.

In practice, it is difficult to alter the wettability of a surface without altering other surface properties. In the past, polymer layers, LB films, SAM, covalent grafting and surfactant adsorption from solution have been employed to control the chemical nature of surfaces in studies of the BCs. All of these modifications have undesired influences on the elasticity and nanoscale roughness of the interface. Thus any observed changes in the BC cannot easily be ascribed to a change in surface energy alone. A potential solution to this problem could be to employ surfaces that exhibit a significant change in charge (i.e. surface energy) as a function of pH or concentration of potential-determining ion. Silica and mica in aqueous solution are unsuitable as the contact angle is not significantly altered by changes in pH. However, pH sensitive SAM and sapphire surfaces may prove suitable for such studies.

A few other problems appear to be of fundamental importance and should be addressed in the near future:

- (i) completing the studies of flow behaviour of simple liquids at soft or patterned interfaces [229, 299, 300].
- (ii) in cases where slip depends on desorption or adsorption of molecules at the interfaces, a thorough experimental study of slip as a function of temperature is necessary [118, 254, 301].
- (iii) MD studies may be applied to the investigation of the local temperature of molecules during the slip process to address if surface cooling of the solvent or frictional heating of molecules occurs as a result of slipping.
- (iv) it is possible that a connection exists between the slip in laminar flows, described in this review, and the slip occurring in low level turbulent flows, upon addition of small quantities of polymers [302]. Existing analogies have not yet been investigated.

8. Conclusions

We have reviewed recent experimental investigations of the hydrodynamic BCs for flows of Newtonian liquids. An increasing number of techniques suitable for the study of flows near interfaces have found evidence for partial slip of simple liquids at solid boundaries. There are discrepancies in the magnitude of slip measured even within each experimental method: results appear to be highly sensitive to the nature of the surface, in a manner that is not yet well defined. Also, experimental measurements of slip frequently give rise to slip lengths that are much larger than those obtained by simulations. It appears likely that the presence of gaseous

films or nanobubbles at the interface may affect a number of published measurements; indeed the largest slip lengths reported can at this stage be explained only by allowing for a gas–liquid interface. A greater understanding of the stability and dynamics of nanobubble formation is required.

We remain convinced of the importance of surface roughness and wettability in determining the BCs. It is however a challenge to separate their influences and isolate them from other important experimental parameters. Roughness may prove to have a dominating influence on slip, though an adequate description of surface roughness and a suitable way of defining the hydrodynamic zero of separation in drainage force measurements are still missing, particularly for surfaces with roughness on scales significantly larger than the molecular scale.

With a greater understanding of the surface properties that influence the BCs, we envisage we will be able to prepare surfaces with known and controlled BCs. Ultimately, we will be able to prepare surfaces with responsive BCs, i.e. surfaces on which the interfacial flow can be altered, perhaps by varying shear rate of the liquid or by tuning other parameters of the surface, such as surface charge. The engineering of such surfaces will be of great interest in a number of applications and in a range of technologies, from lubrication to microfluidics.

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